SEBIN DEVASIA

Male | 34 | Married
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EXPERIENCE

- **Assistant Professor** *01/2025 - Till Now*
  PSG Institute of Technology and Applied Research, Coimbatore, India [🌐]
  - Conduct full-time research, write grant proposals and engage in challenging research problems.
- **Assistant Professor** *10/2023 - 01/2025*
  PSG Institute of Advanced Studies, Coimbatore, India [🌐]
  - PI of Laboratory for Atomistic Simulations and Energy materials Research ($LASE_mR$ group)
  - DFT calculations using Quantum ESPRESSO and generative models for materials design in photocatalysis, thermoelectrics and photovoltaics.
  - Synthesis of lead-free halides, chalcogenides and chalcohalide light absorbers.
  - Guiding masters project students
- **Doctoral researcher** *01/2019 - 05/2023*
  Universidad Autónoma de Nuevo León (UANL), Monterrey, Mexico [🌐]
  - Deposition of $Cs_3Bi_2I_9$ thin films using ultrasonic spray deposition method.
  - Structural, optical, electronic, morphological characterization of thin films.
  - Fabrication of photodetectors and photovoltaic devices and analysis.
  - DFT calculations for structure optimization, electronic band structure and density of states of bismuth halides using VASP.
- **Junior/Senior Research Fellow, DAE-BRNS Project** *07/2015 - 03/2018*
  Union Christian College, Aluva, India [🌐]
  - Investigation of highly conducting Al-doped ZnO (n-Type) and P or N doped ZnO (p-Type) thin films by spray pyrolysis.
  - Chemical methods to grow vertically aligned ZnO nanorods on spray-deposited seed layers.
  - Fabrication of translucent n-ZnO/p-ZnO homojunction diodes.

EDUCATION

- **Ph.D. in Materials Engineering [🌐]** *01/2019 - 05/2023*
  Universidad Autónoma de Nuevo León (UANL) Monterrey, Mexico
  - Thesis:"*Ultrasonic spray deposition of perovskite-inspired lead-free cesium bismuth iodide thin films for optoelectronics*"
  - Grade: Summa cum laude
- **M.Sc. Physics [🌐]** *06/2012 - 08/2014*
  National Institute of Technology (NIT) Tiruchirappalli Tiruchirappalli, India
  - Result: 83.7 %
- **B.Sc. Physics[🌐]** *05/2009 - 04/2012*
  Mahatma Gandhi University Kottayam (Deva Matha College Kuravilangad) Kottayam, India
  - Result: 87.25 %

SKILLS

- **Computational software/Methods:** VASP-DFT, Quantum ESPRESSO-DFT/MD, Machine Learning
- **Generative models and tools for computational studies:** MACE, FairChem models, MatterSim, MatterGen, Orb Materials, SOD, BoltzTrap2 code
- **High Performance Computing:** PARAM Rudra (IUAC, Delhi), PARAM Seva (IIT Palakkad), PARAM Vidya (IIT Hyderabad)
- **Scientific illustration:** Blender, Adobe illustrator & Photoshop
- **Programming Languages:** Python (intermediate), C/C++
- **Material & Device characterization:** XRD, XPS, Raman, SEM, TEM, UV-Vis, Photoluminescence, EDX, FT-IR, Photocurrent response, JV characteristics
- **Synthesis techniques:** Thermal evaporation, Spray deposition (ultrasonic and air-assisted spray), Chemical bath deposition, Aqueous chemical growth, Sol-gel method

TRAINING/WORKSHOPS

- **One month course on "Quantum Computing"** *03 - 25 May 2025*
  C-DAC, Hyderabad & IIT, Roorkee with the support of MeitY, Govt of India
- **5-Day Faculty Development Program (FDP) on "Generative AI"** *10 - 14 May 2025*
  Electronics & ICT Academy, C-DAC Hyderabad, and MeitY, Govt of India
- **Online Hands-on Training on DFT Modelling at Molecular Level using Gaussian** *18 – 24 March 2025*
  Centre for Advanced Computational Research, New Delhi
- **International School on Modelling and Simulations of Materials** *20 - 22 January 2025*
  JNCASR, Bangalore, India
- **PWTK-2024: PWTK scripting environment - Online** *20 – 24 May 2024*
  MaX CoE in collaboration with NCC Slovenia
- **Hands-on-Training on Density Functional Theory Modelling of Materials** *22 - 28 October 2023*
  Centre for Advanced Computational Research, New Delhi

PUBLICATIONS

Google scholar

[J.1] S. Devasia, et al. (2024). **In situ grown Bi_2S_3 nanorods in $Cs_3Bi_2I_9$ thin films as broadband self-driven photodetector with improved photostability**. In *Optical Materials*, 152, 115532. DOI: 10.1016/j.optmat.2024.115532

[J.2] S. Devasia, et al. (2023). **Ultrasonically sprayed $Cs_3Bi_2I_9$ thin film based self-powered photodetector**. *Mater. Chem. Phys.*, 296, pp. 127295. DOI: 10.1016/j.matchemphys.2023.127295

[J.3] S. Devasia, et al. (2021). ***In situ* crystallization of 0D perovskite derivative $Cs_3Bi_2I_9$ thin films via ultrasonic spray**. *J. Alloys Compd.*, 893, pp. 162294. DOI: 10.1016/j.jallcom.2021.162294

INVITED TALKS

1. *From atomic potentials to autonomous discovery in energy materials*, International Conference on Advanced Materials for Innovation and Sustainability (ICAMIS 2026), MES Asmabi College, Thrissur, Kerala, India, 3 February 2026
2. *Beginner's workshop on MedeA-VASP atomistic simulation package for solar cell materials*, Talleres LIFYCS 2021, Instituto de Energías Renovables, Universidad Nacional Autónoma de México (UNAM), Mexico, 9 December 2021
3. *(Not Just) Making Things Smaller- Nanoscience and nanotechnology*, KMEA engineering college, Edathala, Kerala, India, 21 January 2021

Honors and Awards

- **Sciences Academies Summer Research Fellowship** *2024*
  Indian Academy of Sciences [🌐]
- **CONACYT Doctoral Fellowship** *Jan 2019*
  CONACYT, Mexico [🌐]

Professional Memberships

- **Perovskite Society of India**, Membership ID: 100 *April 2023 - Present*

Certifications

- **IIT Madras SWAYAM Plus:** AI for Chemistry *Ongoing*
- **LangChain Academy** Project: Deep Agents *Jan 2026*
- **NPTEL:** Machine Learning for Core Engineering Disciplines *2025*
- **NPTEL:** Programming, data structures and algorithms using python *2017*

Sanctioned Projects

1. **UGC-DAE CSR: Collaborative Research Project (CRS/2023-24/04/1315)**
   Title: An Integrated Approach in Advancement of Quaternary Oxychalcogenide Synthesis for Thermoelectric Application Centre: Kalpakkam, Role: Co-PI, PI: Dr. Geetha Priyadarshini, Duration: 2024-2027, Budget: 12 L

Additional Information

Languages:
English (Fluent),
Malayalam (Native language),
Hindi (School education),
Tamil (intermediate),
Spanish (intermediate)

Interests outside work: Graphical Illustrations, Coding, Soccer, Art, Cooking, Hiking, Reading

References

1. **Dr. E. I. Anila**
   Professor
   Christ University, Bangalore, India
   Email: anilaei@gmail.com
   Mobile: +91 88910 08189
   Relationship: Project Supervisor (3 years)

2. **Dr. Bindu Krishnan**
   Professor, Facultad de Ingeniería Mecánica y Eléctrica (FIME)
   Universidad Autónoma de Nuevo León (UANL), Mexico
   Email: kbindu_k@yahoo.com
   Mobile: +52 81168 80733
   Relationship: PhD Thesis Advisor (4.5 years)